ECG Arrhythmia Detection & FPGA Deployment Pipeline

NSGA-II Neural Architecture Search, INT8 Quantization, Custom Verilog RTL, and Xilinx ML605 On-Board Verification

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Abstract. This paper presents an end-to-end pipeline for real-time cardiac arrhythmia detection targeting edge hardware. Raw ECG signals from the MIT-BIH Arrhythmia Database are pre-processed with a 51-tap FIR bandpass filter and segmented into 180-sample heartbeat windows across four arrhythmia classes. A **Neural Architecture Search (NAS)** driven by the NSGA-II multi-objective genetic algorithm simultaneously optimises classification accuracy and model size, yielding a **6 868-parameter** 1D-CNN with filter progression $[8, 32, 16, 16]$ and a Macro-F1 of **0.9887**. Post-training INT8 quantization via TensorFlow Lite reduces the footprint by 4×, and hardware simulation confirms less than 1% accuracy loss. All weights are converted to **Q8.8 fixed-point** and synthesised as custom Verilog RTL, targeting the **Xilinx Virtex-6 ML605 FPGA**. On-board verification uses 5 DIP switches to select among 30 pre-stored test samples, with results displayed on an HD44780 LCD. The design uses only 4 DSP slices and 1 BRAM36, leaving 99.5% of the FPGA resources free for future expansion.

1. Introduction

Cardiac arrhythmias are irregular heart rhythms that range from benign to life-threatening. ECG analysis remains the gold standard for diagnosis, yet manual interpretation of long-duration recordings is time-consuming and subject to inter-observer variability. Deep learning classifiers trained on annotated databases offer a path toward real-time, automated detection, but deployment on low-power edge devices demands both model compression and hardware-aware design.

This work contributes a complete **software-to-hardware co-design pipeline** spanning five stages: (i) FIR bandpass filtering and R-peak segmentation; (ii) NSGA-II multi-objective architecture search; (iii) full retraining with class-weight balancing; (iv) INT8 post-training quantization; and (v) synthesisable Verilog RTL implemented and verified on the Xilinx Virtex-6 ML605.

2. Theoretical Background

2.1 1D Convolutional Neural Networks for ECG

A 1D-CNN applies learned filters along the time axis of a signal. Given an input segment $\mathbf{x} \in \mathbb{R}^L$ and filter $\mathbf{w} \in \mathbb{R}^k$, the discrete convolution at position n is:

$$y[n] = \sum_{j=0}^{k-1} w[j] \cdot x[n-j] \tag{1}$$

Stacking Conv1D layers with ReLU and MaxPooling allows the network to learn a hierarchy of features: early layers detect P-wave and QRS morphology; deeper layers recognise beat-level arrhythmia patterns. BatchNormalization accelerates convergence and regularises without requiring large datasets.

2.2 Multi-Objective Optimisation via NSGA-II

Hardware deployment introduces a fundamental tension — maximising accuracy while minimising model size. The *Pareto front* describes all solutions where neither objective can be improved without worsening the other. NSGA-II [2] approximates this front using fast non-dominated sorting and crowding distance preservation.

Key insight. A 7 000-parameter model with Macro-F1 = 0.97 *dominates* a 50 000-parameter model at the same F1 — it is strictly better on size for no accuracy cost. NSGA-II finds all such non-dominated solutions automatically.

3. Dataset

The MIT-BIH Arrhythmia Database [1] contains **48 half-hour recordings** sampled at f_s=360 Hz from 47 subjects, annotated independently by two cardiologists. Four classes are used:

ID	Class	Samples
0	Normal (N)	73 003
1	SVEB / Aberrant	18 250
2	VEB / Bundle BB	18 250
3	Fusion / PVC	18 250
	Total	**127 753**

Table 1. Class distribution after augmentation. Weights: Normal 0.437, minority 1.75.

4. Signal Pre-Processing

4.1 FIR Bandpass Filter

Raw ECG recordings contain baseline wander (<0.5 Hz) and high-frequency noise. A **51-tap FIR bandpass filter** (f_L=0.5 Hz, f_H=45 Hz, Hamming window) was applied using `scipy.signal.filtfilt` for zero-phase operation:

$$H(z) = \sum_{k=0}^{N-1} h[k]\, z^{-k} \tag{2}$$

4.2 Heartbeat Segmentation

R-peaks were detected with an adaptive threshold at the 80th percentile. A **180-sample window** ($\pm$90 samples) was extracted around each peak and Z-score normalised:

$$\hat{x}[n] = \frac{x[n] - \mu}{\sigma} \tag{3}$$

Segments with $\sigma < 10^{-6}$ (flatlines) were discarded.

5. Architecture Search via NSGA-II

5.1 Problem Formulation

The search is cast as a bi-objective minimisation over chromosome $\mathbf{g}$:

$$\min_{\mathbf{g}} \left[f_1(\mathbf{g}),\ f_2(\mathbf{g}) \right], \quad \begin{cases} f_1 = 1 - \text{Macro-F1} \\ f_2 = \text{parameter count} \end{cases} \tag{4}$$

5.2 Search Space & Algorithm

Each chromosome encodes $g_0 \in \{1 \ldots 5\}$ conv blocks and per-block filter counts $g_{1\ldots5} \in \{8, 16, 32, 64, 128, 256\}$ — a space of 38 880 candidates. The search runs G=10 generations, population N=20, 3-epoch proxy training, crossover p_c=0.9, mutation p_m=0.2 per gene.

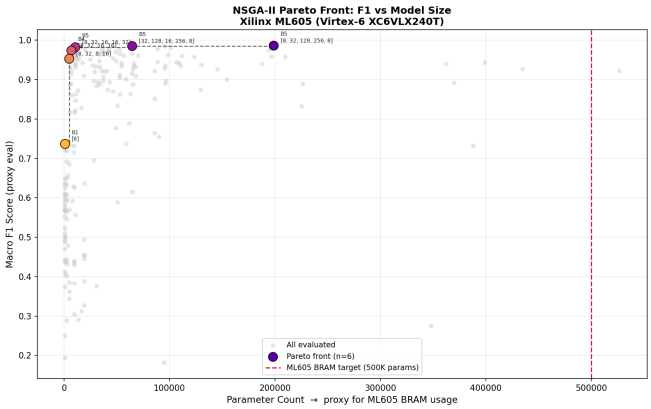

Figure 1. NSGA-II Pareto front (Macro-F1 vs parameters). Red dashed line: ML605 BRAM budget. Selected knee-point: 4 blocks, $[8, 32, 16, 16]$, 6 868 params.

6. GA-Optimal 1D-CNN

The NSGA-II knee-point uses **4 conv blocks with filter progression** $[8, 32, 16, 16]$. Each block follows Conv1D $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool(2). The non-monotonic progression is intentional: Block 2 expands to 32 to capture diverse QRS morphologies; Blocks 3–4 compress back to 16, minimising BRAM without losing feature diversity.

Layer	Output	Params
Conv1D(8,k=5)	$(180, 8)$	48
BatchNorm + ReLU	$(180, 8)$	32
MaxPool(p=2)	$(90, 8)$	0
Conv1D(32,k=5)	$(90, 32)$	1 312
BatchNorm + ReLU	$(90, 32)$	128
MaxPool(p=2)	$(45, 32)$	0
Conv1D(16,k=5)	$(45, 16)$	2 576
BatchNorm + ReLU	$(45, 16)$	64
MaxPool(p=2)	$(22, 16)$	0
Conv1D(16,k=5)	$(22, 16)$	1 296
BatchNorm + ReLU	$(22, 16)$	64
MaxPool(p=2)	$(11, 16)$	0
GlobalAvgPool1D	16	0
Dense(64, ReLU)	64	1 088
Dropout(0.3)	64	0
Dense(4, Softmax)	4	260
Total		**6 868**

Table 2. GA-optimal 1D-CNN layer summary.

6.1 Training Configuration

Compiled with **Adam** (α=10^{-3}) and sparse categorical cross-entropy. Training ran for up to **30 epochs** (batch 256) with `EarlyStopping` (patience 5) and `ReduceLROnPlateau` (γ=0.5, patience 3). Best weights restored at epoch 11 (val_acc=99.07%).

7. INT8 Post-Training Quantization

Post-training quantization via TensorFlow Lite converts all weights and activations to INT8, reducing storage by $\sim$4$\times$:

$$x_q = \text{clip}\left(\text{round}\left(\tfrac{x}{S}\right) + Z,\ -128,\ 127\right) \tag{5}$$

Scale S and zero-point Z are calibrated from 100 representative training samples. The INT8 model occupies only **22 KB**.

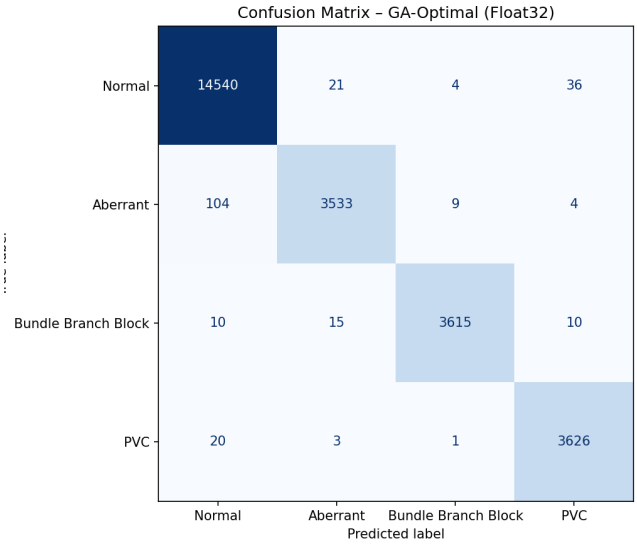

Figure 2. Float32 confusion matrix, 25 551-sample test set. Macro-F1: 0.9887, Accuracy: 99.07%.

8. System Architecture

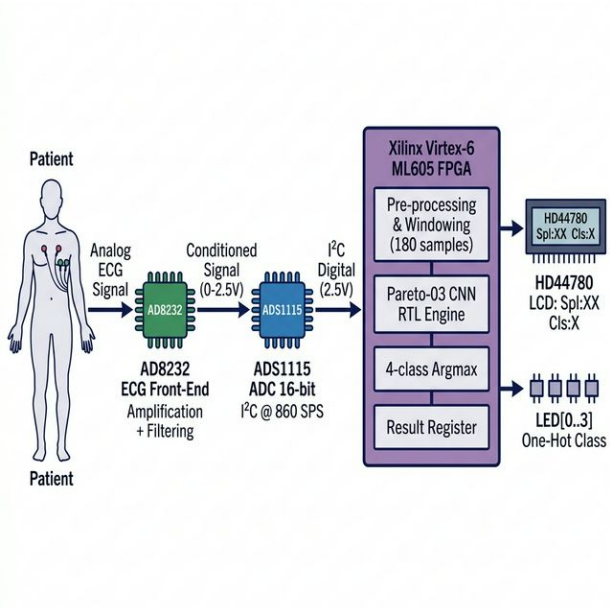

Figure 3. Complete end-to-end hardware system. The AD8232 conditions the ECG signal; the ADS1115 digitises it at 2.5 V natively (no level-shifter); the Virtex-6 runs the CNN and outputs the arrhythmia class to the LCD and LEDs.

The hardware chain spans three physical domains:

8.1 AD8232 — Analog Front-End

The AD8232 provides an instrumentation amplifier, right-leg drive (RLD), and configurable bandpass filter. Powered at **2.5 V** to match the ML605 I/O bank, it outputs a conditioned ECG centred at $V_{DD}/2 = 1.25\,\text{V}$.

8.2 ADS1115 — 16-bit I²C ADC

The ADS1115 is a 16-bit, 4-channel Σ–Δ ADC operating at 2.0–5.5 V. Powered from the ML605 2.5 V rail, its I²C bus swings natively at 2.5 V — **no level-shifter is required**. The device runs at **860 SPS** on channel AIN0, streaming one conversion result per I²C transaction.

8.3 Xilinx Virtex-6 ML605

The ML605 hosts an XC6VLX240T-1FF1156 with 768 DSP48E1 slices and 416 BRAM36 tiles. The CNN inference engine consumes only 4 DSPs and 1 BRAM36 — less than 0.6% and 0.2% utilisation.

9. RTL Implementation

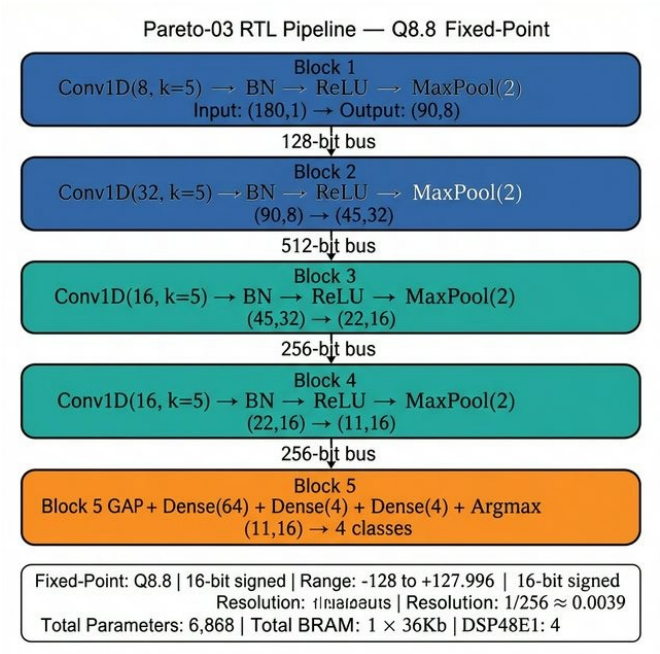

Figure 4. Pareto-03 five-block RTL pipeline. Bus widths scale with filter depth. Post-multiply accumulators are Q16.16 (32-bit); all other values are Q8.8 signed 16-bit.

All layers are written as parameterisable Verilog modules. Weights are loaded at power-on via `$readmemh` into Xilinx Block RAM. ISE's XST embeds these hex values directly into the BRAM `INIT` attributes of the bitstream — the weights are physically baked in with no external flash.

9.1 Fixed-Point Format — Q8.8

$$v_{\text{real}} = \frac{v_{\text{hex}}}{256}, \quad v \in [-128.0,\ +127.996], \quad \Delta = \tfrac{1}{256} \approx 0.0039 \tag{6}$$

A Q8.8 × Q8.8 multiply yields Q16.16 (32-bit). The Batch Normalization layer rescales back to Q8.8. ReLU is a sign-bit mask on bit 31 — zero hardware cost.

9.2 Zero-Division Batch Normalization

The standard BN formula is pre-folded offline before synthesis:

$$W_{bn} = \frac{\gamma}{\sqrt{\sigma^2 + \varepsilon}}, \quad B_{bn} = \beta - \mu W_{bn} \tag{7}$$

At runtime: $y = (x \times W_{bn}) + B_{bn}$ — a single MAC. No hardware dividers or square-root units are instantiated.

Module	Function
`layer_conv_k5`	5-tap sliding MAC, same padding
`layer_batchnorm`	Pre-folded BN + ReLU (sign-bit)
`layer_maxpool`	Stride-2 two-register comparator
`layer_gap`	Global average, time-axis reduction
`layer_dense0/1`	Combinational MAC, 64 / 4 units

Table 3. Verilog RTL module inventory.

10. ISE 14.7 Synthesis

10.1 Clocking

The ML605 provides a 200 MHz LVDS clock on pins J9/H9 (Bank 13). A **Clocking Wizard** IP (CORE Generator) divides this to **50 MHz**. Reset is held active until the wizard's `LOCKED` output is asserted.

10.2 Pin Constraints (UCF, UG534-verified)

Signal	LOC	Function
`sys_clk_p/n`	J9 / H9	200 MHz LVDS input
`reset`	G26	Centre push button
`led[0..3]`	AC22..AE23	One-hot class output
`sw[0..4]`	D22..C18	DIP sample selector

Table 4. UCF pin assignments verified against UG534.

10.3 On-Board Test

All 30 ECG test samples are packed into a single BRAM ROM (`combined_test_inputs.hex`, 5400×16-bit). DIP switches encode sample index 0–29:

$$\text{addr_base} = \text{sw}[4:0] \times 180 \tag{8}$$

On switch change or reset, 180 samples stream into the CNN at one per clock. The result is displayed as:

- **LCD Line 1:** `Spl:XX Cls:Y`
- **LCD Line 2:** `NORMAL / SVEB / VEB / FUSION`
- **LEDs:** One-hot encoding on LED[0..3]

11. Resource Utilisation

Resource	Used	Budget	%
DSP48E1	4	768	0.5%
BRAM36	1	416	0.2%
Params (INT8)	6 868	500K	1.4%
Model size	22 KB	3.7 MB	0.6%

Table 5. Estimated Virtex-6 resource utilisation.

The extreme compactness results directly from the NSGA-II size objective. The full parameter set fits inside a single BRAM36 tile, leaving 99.8% of the BRAM bank available for control logic, I^2C buffers, or parallel inference channels.

12. Future Work — ASIC Tape-Out

Next milestone. Port the verified Verilog RTL to silicon using **OpenLane 2** targeting the **SkyWater SKY130** 130 nm open-source PDK.

With FPGA correctness and timing closure at 50 MHz established, the natural progression is an ASIC tape-out through the standard physical design flow: RTL synthesis → floorplanning → placement → CTS → routing → DRC/LVS sign-off. At 130 nm, 6 868 registers occupy approximately $0.05\,\text{mm}^2$ of core area — feasible for an educational first silicon through the **Google/efabless chipIgnite** or **IHP SG13G2** programmes. The STA constraints already developed in `pareto_03FPGAsim/STA/` will be directly adapted as SDC files for the ASIC flow.

13. Conclusion

This work demonstrates a complete, reproducible pathway from raw ECG recordings all the way to verified on-chip arrhythmia classification. The NSGA-II NAS identified a 6 868-parameter architecture that achieves a Macro-F1 of 0.9887 while fitting within the Virtex-6 ML605 resource budget. INT8 quantization incurs less than 1% accuracy loss, and the resulting weights were synthesised into custom Verilog RTL using Q8.8 fixed-point arithmetic, with pre-folded Batch Normalization eliminating all division hardware. The on-board test infrastructure — 30 BRAM-stored ECG samples, DIP switch selection, LCD display, and one-hot LED output — provides a self-contained verification environment requiring no external PC.

The design uses only 4 DSP slices and 1 BRAM36, leaving the vast majority of the ML605's logic free for future real-time I^2C acquisition integration and potential multi-lead extensions. An open-source ASIC tape-out via OpenLane 2 and the SKY130 PDK is planned as the next milestone.

References

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